Blaqks One — Technology Stack

The tools behind the Blaqks One real-estate CRM platform: what each one is, why we use it and the problem it solves.

In use running today **At launch** needed for hosting and public sign-up **Later** added with the named feature

Frontend

React 18 — In use

Why: Widely used component library for building screens from reusable pieces

Solves: Keeps a large CRM interface consistent and maintainable

React Router — In use

Why: Standard page navigation for React

Solves: Each screen has its own address and survives a reload

Vite — In use

Why: Very fast development server and production build tool

Solves: Instant updates while building; small, fast files for users

JavaScript and CSS — In use

Why: Plain, well-known languages with light and dark themes

Solves: No heavy framework lock-in; easy to hire for

JSZip — In use

Why: Small library for zip files

Solves: Builds Excel workbooks and backups without a large spreadsheet library

Backend

Rust — In use

Why: Fast, memory-safe language that catches many bugs before release

Solves: Reliable server that handles money and permissions safely

Axum and Tokio — In use

Why: Modern Rust web framework and asynchronous runtime

Solves: Serves many users at once with little hardware

SQLx — In use

Why: Database library that checks queries and runs migrations

Solves: Safe database access and versioned database changes

Argon2 — In use

Why: Recommended password-hashing method

Solves: Stolen database copies cannot reveal passwords

Rustls — In use

Why: Modern, memory-safe encryption library

Solves: Encrypted connections without OpenSSL vulnerabilities

Data and storage

PostgreSQL 17 — In use

Why: Proven, open-source database with strong transactions and integrity rules

Solves: One trusted source of truth; each company's data stays separate and consistent

Docker and Docker Compose — In use

Why: Runs PostgreSQL the same way on every machine

Solves: Identical development setup; disposable databases for tests

Managed PostgreSQL in a UAE region — At launch

Why: Hosted database with automatic backups, updates and failover

Solves: No manual database operations; meets UAE data-location expectations

Object storage in a UAE region (S3-compatible) — At launch

Why: Durable storage for property photos, floor plans and agreement evidence

Solves: Media off the database server, cheap to store and served through the CDN

Valkey (open-source Redis) — Later

Why: In-memory cache and fast counters

Solves: Fast public listing pages and real-time features when traffic grows

Hosting and traffic

Cloud provider with a UAE region — At launch

Why: For example AWS (UAE) or Azure (UAE North)

Solves: Client data stays in or near the UAE under its data-protection law

Cloudflare CDN — At launch

Why: Global network that serves files and images from near the user

Solves: Fast website and photos; blocks attacks and bad traffic before it reaches us

Cloudflare Turnstile — At launch

Why: Privacy-friendly bot check on sign-up and sign-in

Solves: Stops fake sign-ups and automated password guessing

Nginx (reverse proxy and gateway) — At launch

Why: Industry-standard entry point in front of our services

Solves: HTTPS, routing `/api/crm`, `/api/public` and `/api/calendar`, rate limits, security headers

Forward proxy for outbound calls — Later

Why: Controls which outside services our servers may call

Solves: Limits and logs calls to WhatsApp, Meta and AI providers

Dedicated API gateway (for example Kong) — Later

Why: API keys, usage plans and billing per company

Solves: Only needed if Blaqks One offers an API to outside partners

Private CRM data is never cached by the CDN or proxy; only public files, photos and approved public listings are cached.

Accounts, messaging and integrations

Transactional email service — At launch

Why: For example Amazon SES or Postmark

Solves: Sign-up verification, invitations and password recovery reach inboxes

WhatsApp Business Platform — Later

Why: Official WhatsApp API

Solves: Send and receive client messages from the CRM, within Meta's rules

Meta (Facebook and Instagram) APIs — Later

Why: Official business-account connections

Solves: Social messages and ad leads arrive in the CRM

UAE PASS — Later

Why: UAE government digital identity and signing

Solves: Strongest proof for cross-company deal agreements

Online payments (for example Stripe) — Later

Why: Subscription billing for companies

Solves: Companies pay for Blaqks One plans online

Google Calendar companion — In use (paused)

Why: Separate small Rust service, kept for a later phase

Solves: Two-way calendar sync when it resumes

Quality, delivery and operations

Node test runner, Cargo test — In use

Why: Unit and API tests for the frontend and backend

Solves: Catches broken rules and calculations before release

Playwright — In use

Why: Real-browser tests of complete user journeys

Solves: Proves screens work end to end, on phone and desktop sizes

Prettier, rustfmt, Clippy, size check — In use

Why: Automatic formatting, code-quality checks and a 150-line file limit

Solves: Consistent, readable code that stays easy to change

cargo-deny, npm audit and Dependabot — In use

Why: Scans dependencies for known security holes and licence problems; opens weekly update pull requests

Solves: Vulnerable or wrongly licensed libraries are caught and updated quickly

Git and GitHub — In use

Why: Version history and code hosting

Solves: Every change is tracked and recoverable

GitHub Actions — In use

Why: Runs all checks automatically on every change

Solves: Nothing is released without passing tests

Error tracking and uptime monitoring — At launch

Why: For example Sentry plus an uptime checker

Solves: We learn about problems before customers report them

Encrypted backups and restore drills — At launch

Why: Regular, tested database and media backups

Solves: Data can be recovered after mistakes or outages

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